

The Fundamental Axiom of LLMs

Computation, Continuation, and Governance

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Abstract

Descriptions such as "the model hallucinated," "the model got confused," or "the model lied" assign an interpretation to generated behavior. They do not, by themselves, identify its generating mechanism. This paper locates interpretation in governance and distinguishes it from computation, decoding, and subsequent control. A reasoning continuation can condition later computation without faithfully reporting the process that produced it. The continuation is not a confession of internal state. Here, "axiom" names a discipline for explanation. The contribution is an explicit account of computational and interpretive boundaries; the propositions record elementary consequences of the stated definitions. Sufficient state makes information persistence explicit. Property-specific identification separates exact determination, identification from an evidence distribution, and finite-sample inference. Semantic conventions must be declared and mechanistic attributions supported. The paper makes no disposition about consciousness, experience, or the computational realization of higher-level functional properties.

1. The explanatory problem

An output is observed. A reader calls it confused, deceptive, hallucinatory, or intentional. The attribution then becomes the explanation: the output occurred because the model was confused, deceptive, hallucinating, or pursuing an intention. The last step introduces a causal claim that the observation and its label do not establish.

The problem is the substitution of interpretation for mechanism. A word may classify an output, express a reader's reaction, summarize a pattern, or assert a property of the computation. Those uses have different evidentiary requirements. Moving between them without acknowledgment lets a familiar word supply a mechanism that has not been identified.

This paper calls the interpretive and evaluative function **governance**. Governance includes the researcher selecting an experimental criterion, the operator diagnosing a trace, the reviewer judging a response, and an automated evaluator assigning a category. This is a functional definition, not a claim that interpretation belongs exclusively to a particular institution or to humans.

Governance may use whatever terminology serves its purpose. If an existing word is given a new operational meaning, the redefinition should be explicit. Evidence for the new definition does not automatically support the additional implications of the ordinary word.

The formulation used here is:

An interpretation of a generated continuation does not, by itself, identify the mechanism that produced it. A mechanistic attribution must identify the computational relation asserted and the evidence supporting that relation.

2. The computational process

2.1. The primitive and its boundary

For an ordinary context-determined forward pass, write

$$h_k = f_\theta(c_k), \quad z_k = G_\theta(h_k).$$

The fixed parameters are θ . The context actually admitted at inference step k is c_k . The hidden vector used for readout is h_k , and the logit vector over vocabulary V is $z_k \in \mathbb{R}^{|V|}$. The mapping G_θ includes the readout operations needed to produce logits; it need not mean only a linear unembedding matrix.

At this boundary, the model primitive is $G_\theta \circ f_\theta$. Decoding is explicit and downstream. Let K_ϕ be the decoder's probability kernel, with configuration ϕ and any relevant decoder state included in Σ_k :

$$x_{k+1} \sim K_\phi(\cdot \mid z_k, \Sigma_k).$$

This includes deterministic selection as a point-mass distribution. Temperature, truncation of the token distribution, and stateful constraints belong to the declared decoding process. The notation does not attribute sampling randomness to the fixed forward mapping.

The context is admitted material, not necessarily the complete transcript. It may contain instructions, retrieved content, prior tokens, summaries, or tool observations. Composition may select, quote, truncate, replace, or reset that material. What was supplied to an interface need not equal what reaches the model.

These conventions follow the author's notation reference and deployment framework [1, 2]. The present paper uses k for inference steps, reserving t for broader operational time. In ordinary autoregressive decoding, each step selects one token and k also indexes token generation. The event-based extension below permits steps without emission. Selected tokens are denoted by x_{k+1} .

2.2. Sufficient inference state

A single readout vector does not generally contain all information needed to continue inference. Let $\Sigma_k \in S$ denote a **sufficient running inference state at a declared step boundary**. Together with the fixed computational configuration and subsequently admitted events, it supports the next transition and emission.

Depending on the computation, this state can include retained context, cached representations, latent feedback, relevant activations, decoder bookkeeping, and execution position. These components need not be independent. Sufficiency does not establish minimality, and no minimal representation is required here. A hidden vector may be a component or a derived quantity of Σ_k .

For the full step, including input admission and any output selection, write

$$\Sigma_{k+1} = U(\Sigma_k, e_k, \omega_k), \quad x_{k+1} = D(\Sigma_k, e_k, \omega_k).$$

The fixed model, decoder, and update configuration are suppressed in U and D . The admitted event e_k includes any external input or intervention relevant to that step; a no-new-input event permits internal computation. The variable ω_k supplies any randomness required to make these maps deterministic. For a step without token emission, D may return a declared no-output symbol. This permits latent computation between token emissions.

The ordinary decoding kernel above is a special case of this event-based description. U is the state-update operator; D names the complete step's emission, without replacing f_θ , G_θ , or K_ϕ at their respective

boundaries. An independently altered cache or latent state must be represented in Σ_k or in an admitted intervention. The original text alone need not reproduce that altered computation.

The engineered control layer C and the context-update process can themselves alter what is admitted. If an account concerns an entire agent or deployed system, its sufficient state must also cover the relevant harness and environment. The notation reference reserves Ω_t for that broader operational state. The inference and operational boundaries should not be conflated.

2.3. Persistence and restart

The restart thought experiment makes the information requirement concrete. Run a forward pass, retain the information needed for continuation, power down the hardware, reconstruct the computational configuration, and resume. A cache can be saved or, where it is derived from retained context and parameters, recomputed. A cache that incorporates an external intervention requires that intervention's effect to be retained or reconstructed as well.

Proposition 1. Conditional replay. For fixed U and D , two executions with the same sufficient initial state and the same admitted-event and randomness sequences produce identical states and emissions at every finite step.

Proof. Their first transitions and emissions have identical arguments. The resulting states are therefore identical. Repeating the argument proves the statement by induction. $\square$

This is a statement about the declared computation, not a promise of bitwise reproducibility across arbitrary hardware or numerical implementations. Reproducing a distribution does not require repeating a particular random draw; reproducing an exact sampled execution does.

Under the declared sufficient-state model, any influence attributed to an earlier event passes through information retained or reconstructed by the resumed computation. Whether the declared state is sufficient for a deployed system is a modeling obligation: omitting relevant information invalidates that application of the replay claim. The requirement concerns information, not the survival of a particular capacitor, register, neuron, or activation value. A representation can be recomputed and still have a causal role.

2.4. Higher-level descriptions

This computational account permits meaningful descriptions of organization within the process. A concept may identify related configurations across contexts; a functional description may identify a repeatable transformation or dependency. Such descriptions need not correspond to one token or one persistent register.

The state definition alone does not validate a particular concept attribution. Nor does implementation by fixed operations invalidate it. The relevant question is what structure the description identifies and what evidence supports the proposed correspondence. Naming the entire mapping establishes a framework for explanation; it does not explain a particular output.

3. Interpretation belongs to governance

Let τ denote a trajectory of the declared computation, including the state and event information relevant to the analysis. Let Π map that trajectory to the observations available to governance:

$$O = \Pi(\tau), \quad \ell = I_\Gamma(O, E).$$

The observation O may include tokens, traces, logs, selected activations, or experimental measurements. The auxiliary evidence E may include known configuration, intervention records, and reference material. The

governance configuration Γ supplies the interpretive criteria, and ℓ is the assigned description. If interpretation uses randomness or changing evaluator state, those belong to the declared governance configuration or its inputs.

Observation, interpretation, and control perform distinct functions. Π specifies what is available; $\mathcal{I}_\Gamma$ assigns a meaning or category; C and U determine how that assignment affects subsequent execution, if it does. Governance is not causally inert: a judgment can become a new input or control decision. Its downstream effect does not retroactively establish its accuracy as a description of the earlier event.

Consider three statements:

1. The output disagrees with a declared reference.
2. Governance labels that disagreement "lying."
3. The model represented a contrary proposition and intentionally misrepresented it.

The first is a comparison. The second is an interpretive convention. The third asserts additional properties of the generating process. Establishing the comparison does not establish those properties. If "lying" has been redefined to mean only reference disagreement, the narrower definition should be stated and conclusions should remain attached to it.

The same discipline applies to "hallucination," "confusion," "introspection," and claims of an internal person or agent. A name can be useful without supplying a causal explanation. Governance should state what it means, where the interpretation occurs, and which part of the claim is experimentally or formally supported.

This also applies to the terminology of this paper. "Governance" has the functional definition given in Section 1. "Sufficient state" has the computational meaning given in Section 2. Neither label imports an additional causal entity.

4. Reasoning continuations are part of the computation

4.1. The continuation is not a confession of internal state

A reasoning continuation can contain first-person statements about a choice, belief, constraint, or intention. Such statements are generated outputs. Their grammatical form and placement in a reasoning trace do not authenticate their content as a report of the state or causal history that produced them.

Nisbett and Wilson examine limits of human verbal reports of cognitive processes [3], a historical precedent for scrutiny of self-report; the computational claims here require their own evidence. In NLP interpretability research, Jacovi and Goldberg distinguish faithful explanations from explanations that seem plausible to a reader [4]. Turpin et al. show that biasing context can change answers without the generated explanation acknowledging that influence [5]. These findings motivate testing the particular relation being asserted.

Three relations are relevant here:

Relation	What it concerns
State dependence	Varying a specified state component or property changes the distribution of the continuation under a declared intervention.
Report fidelity	Its interpreted content accurately describes a specified property of that computation or its history.
Subsequent influence	Varying retained or re-entered text changes later computation under a declared comparison.

The intervention must specify what is varied, what is held fixed, and which states or operations are admissible. Deterministic changes are included as changes between point-mass distributions. State dependence does not establish report fidelity. Subsequent influence does not establish it either. A generated statement can be informative, inaccurate, partly faithful, or irrelevant to a particular causal question. The relation between the statement and its claimed referent requires evidence.

The argument concerns the actual generated reasoning material available in a specified system. A separately generated summary of that material is an additional output whose evidentiary status must be considered separately.

4.2. A token-level control surface

When generated reasoning tokens are retained or re-entered, they become part of the conditioning for subsequent steps. Lanham et al. investigate this dependence through interventions including early termination, inserted mistakes, and paraphrasing, finding that reliance on CoT varies across tasks and models [6]. Inclusion in context creates a possible route of influence; a particular effect still requires evidence.

The author's phrase **token-level control surface** describes that prospective role [7]. Changing the content or placement of retained text can change subsequent response probabilities. "Control" here means causal influence; it does not imply reliable control, granted authority, or a guaranteed result.

An illustrative sentence is: "The user probably means Q." Once retained, that sentence can favor a continuation organized around Q. It may help establish a trajectory that treats Q as the task. This prospective effect is separate from whether the sentence accurately reports why the preceding computation produced it.

Likewise, retained text that names a constraint can increase or decrease the probability of later responses satisfying it. Repeated justifications and purported corrections can change which answers or formulations recur. These are possible effects to investigate in a specified setting. They require neither a complete nor a faithful report of the text's production.

CoT is not the only route of cross-step influence: retained or recomputed representations may also contribute. The claim is that the token path can participate, not that it exhausts the running state.

4.3. Explanation and participation

A simple abstract example separates the relations. Let the state contain a bit $b \in \{0, 1\}$ and a response-mode flag. The process emits the sentence "The retained bit is zero" for either value of b . Re-entering that sentence changes the response-mode flag and therefore changes the subsequent output format.

The sentence is causally effective in both cases. Its statement about b is accurate in one case and inaccurate in the other. Causal participation therefore does not entail descriptive accuracy. A variant emits "The retained bit is one" when $b=0$ and "The retained bit is zero" when $b=1$: its report depends on the named bit yet misreports it in both cases. When the inverted reporting rule is known, the bit is exactly recoverable from the sentence although the sentence remains false under its literal interpretation. Recovering a property and accurately asserting it are distinct relations. These are logical examples, not proposed transformer implementations.

A post-hoc explanation adds a further distinction. It is generated after the original event, often under changed context. It can direct attention to relevant evidence or to an appealing but non-causal account. Its usefulness must be assessed against the question being investigated. A reason-shaped continuation is not authenticated causal reconstruction.

The deployment framework makes this point for agent diagnostics [2, Section 15]. Here it also applies within reasoning generation: a continuation may be simultaneously an observed artifact, an apparent account of prior computation, and an input to subsequent computation.

5. Evidence and the property being claimed

5.1. Hint influence and acknowledgment

Chen et al. compare questions with and without hints, examine changes toward the hinted answer, adjust for random answer changes, and measure acknowledgment in CoT. They find frequent non-acknowledgment in the studied settings [8, Section 2].

This provides a positive example of a supported attribution. Define "hint influence" as a change in answer distribution caused by inserting the hint. When insertion is the only systematic difference, configuration is held fixed, and sampling variation is accounted for, evidence of that change supports attributing an effect to the input intervention. The declared relation is an input-to-answer dependency. One changed answer pair alone does not establish a distributional effect. Chen et al.'s noise adjustment also assumes a model of random answer changes; its applicability is part of the inference.

Comparing that effect with acknowledgment tests reporting of the influence. It does not identify every internal route producing either the answer or the omission. Intentional concealment is an additional claim. The authors' tasks need not require extended CoT, limiting generalization [8, Sections 1 and 7]. An acknowledgment-based faithfulness score should retain its operational meaning.

5.2. Preserving a useful channel

The multi-author position paper on CoT monitorability argues that readable reasoning can provide useful evidence and should be studied and preserved where beneficial. It explicitly acknowledges incomplete representation and distinguishes computational use of CoT from guarantees of human interpretability or reliable monitoring [9, Sections 1 and 3].

That position is compatible with the present argument. A monitoring channel can be useful without reconstructing the entire state or authenticating every self-description. Its value can be established through its ability to predict or detect specified outcomes. Its use as evidence about a particular internal mechanism requires the corresponding support.

Baker et al. include CoT-monitor scores in the model's training reward. They report improved outcomes under lower optimization pressure, and continued reward hacking with less revealing CoT under stronger

pressure [10]. This is evidence about particular training conditions. It does not imply that every imperfect monitor, or every optimization of its labels, produces that outcome.

5.3. Internal interventions

Lindsey explicitly defines introspective awareness using accuracy, grounding, internality, and metacognitive representation [11, "Defining Introspection"]. Accuracy concerns the report's correspondence to the state; grounding concerns causal dependence on what is reported; internality excludes inference solely from prior sampled outputs. The final criterion requires an internal representation of the state as such, beyond directly expressing its content. These are the source's operational commitments; this paper does not require adopting its terminology.

Lindsey supplies interventions and controls, while stating that the experiments do not pin down a specific mechanism and do not directly demonstrate metacognitive representations. Declaring criteria, supporting them experimentally, and identifying their implementation are distinct steps. The evidence should be credited at the level supported.

Vogel adapts the concept-injection approach to Qwen: steering during prefill, retaining the resulting cache, removing steering, and inspecting later response probabilities [12]. Visible text can remain fixed while retained computational state changes. In this notation, the executions have different Σ_k despite identical text.

This is a second positive example: with other relevant conditions controlled, the cache intervention can establish a state-to-report dependency beyond differences in the visible token history. It does not alone distinguish tracking the claimed property from a direct bias on reporting. The route by which a report is produced matters when that route is part of the claim. An "internality" criterion is therefore relevant to the specified introspection account; it is not a universal requirement for every mechanistic explanation.

5.4. What the available evidence can determine

The relevant question is often whether evidence determines a particular property, not whether it reconstructs the entire state. A non-injective observation map may still preserve the property of interest.

Let $\mathcal{A}$ be a declared set of admissible scenarios and $P: \mathcal{A} \rightarrow \{0, 1\}$ the specified property. A scenario must contain what makes P well-defined: a state or trajectory may suffice, while a counterfactual claim also requires the relevant causal model and intervention semantics.

The property, admissible scenarios, and reasons for excluding alternatives must be stated as premises of the inference. Exclusions require justification independent of the desired conclusion; a scenario cannot be removed merely because it defeats identification. Revising those premises revises the scope of the conclusion.

Let J collect all evidence admitted to the inference, including observations and auxiliary evidence:

$$J(a) = (O(a), E(a)), \quad a \in \mathcal{A}.$$

For a trajectory-based analysis, $O(a) = \Pi(\tau(a))$.

Proposition 2. Exact identification of a specified property. There exists a function d on $J(\mathcal{A})$ satisfying $d(J(a)) = P(a)$ for every $a \in \mathcal{A}$ if and only if

$$J(a) = J(a') \implies P(a) = P(a') \quad (a, a' \in \mathcal{A}).$$

Proof. Necessity follows because a function takes the same value on equal arguments. For sufficiency, assign each evidence value the common value of P among scenarios producing it. The displayed condition makes that assignment well-defined. $\square$

To establish underdetermination of P , exhibit admissible scenarios with identical evidence and different values of P . Merely observing that some state information was lost is insufficient. The criterion is set-theoretic; it does not supply an efficient identification procedure or validate the predicate's interpretation.

For example, if a state is (b, u) and its observation is b , the value of b is recoverable even when u is not. Conversely, the identical sentences in Section 4.3 do not determine whether the retained bit is zero when those sentences are the only evidence available. Recording the bit would change the evidence and resolve that particular ambiguity.

Proposition 2 concerns exact determination from the admitted evidence. A scenario can include a realized random execution; the criterion is not restricted to models that generate deterministically.

5.5. Evidence distributions and finite samples

For statistical identification, let $\mathcal{M}$ be a declared class of mechanism hypotheses. Under a fixed experimental design, each $M \in \mathcal{M}$ induces an evidence law ν_M . Let $P_{\mathcal{M}}(M) \in \{0, 1\}$ be the property at this level. It is identifiable from the evidence law precisely when

$$\nu_M = \nu_{M'} \implies P_{\mathcal{M}}(M) = P_{\mathcal{M}}(M') \quad (M, M' \in \mathcal{M}).$$

This is the same elementary factorization criterion, now applied to the map from a hypothesis to its full evidence distribution. It identifies the property without requiring identification of the entire mechanism. The experimental design, observation process, nuisance assumptions, and hypothesis exclusions are part of the declaration.

If two hypotheses with different property values induce the same full evidence law, every test based solely on that evidence has the same rejection probability under both hypotheses. For a test of one such hypothesis against the other, power equals size.

The declared design may include adaptive or feedback policies, including training and control. The evidence law then covers the resulting experiment. An identification claim applies to that design; changes outside its scope require renewed justification.

Identification from a law does not mean that one realized observation determines the property. For example, let M_0 and M_1 produce Bernoulli observations with probabilities 1/4 and 3/4, respectively, and set $P_{\mathcal{M}}(M_i) = i$. Their laws differ, so the property is identifiable from the law. Yet either hypothesis can produce either observation. Under independent repeated sampling, every finite binary sequence also has positive probability under both hypotheses.

Finite samples can nevertheless support useful estimates, tests, or likelihood comparisons. Their uncertainty depends on the sampling assumptions and available evidence. Identifiability alone supplies neither a sample-size guarantee nor a calibrated decision rule. Conversely, unresolved alternatives do not make observations useless: they limit which conclusions the evidence supports.

6. The obligation of a mechanistic account

A mechanistic explanation should make three things explicit:

1. **The observation or intervention.** State what was recorded, varied, held fixed, or reconstructed, at which computational boundary.
2. **The interpretation.** Define the property or category assigned by governance. Acknowledge operational redefinitions and separate them from the implications of ordinary language.
3. **The supporting relation.** Identify the computational dependency asserted and the evidence that supports it over the stated domain. Distinguish causal influence, descriptive accuracy, prediction, and exact identification.

The obligation is proportional to the claim. Diagnosing a parser rejection may require logs and a known rule. Establishing that a report tracks an internal intervention may require controlled comparisons. Attributing a functional process requires evidence for the organization named. No complete inventory of every activation is demanded merely to explain one relevant dependency.

The paper makes no disposition about consciousness, subjective experience, or whether computation can instantiate concepts, planning, or introspection. It specifies neither a training method nor an engineering construction. Those questions are not resolved by adopting or rejecting a colloquial label.

The same discipline applies to architectural work. CKN states properties of declared dynamics and their relation to specified task operation [13]. Any application of those properties to a trained model requires the relevant correspondence to be established. A vocabulary of privilege, like a vocabulary of cognition, does not supply that correspondence by itself.

The fundamental axiom requires an account of the generating computation and of the interpretive step. Governance may name what it observes. When that name is offered as an explanation, the mechanism and the supporting inference must be made explicit.

References

- [1] Alioto, J. P. (2026). [Notation Reference: A working notation for AI agent deployment in coupled systems](#). Version 1.0, April.
- [2] Alioto, J. P. (2026). [Agent Deployment into Coupled Systems: A Working Framework](#). Version 1.0, April. Especially Sections 3, 4, 15, and 26.
- [3] Nisbett, R. E., and Wilson, T. D. (1977). [Telling More Than We Can Know: Verbal Reports on Mental Processes](#). *Psychological Review*, 84(3), 231-259.
- [4] Jacovi, A., and Goldberg, Y. (2020). [Towards Faithfully Interpretable NLP Systems: How Should We Define and Evaluate Faithfulness?](#). *Proceedings of ACL*, pp. 4198-4205.
- [5] Turpin, M., Michael, J., Perez, E., and Bowman, S. R. (2023). [Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting](#). *NeurIPS 2023*; arXiv:2305.04388.
- [6] Lanham, T., Chen, A., Radhakrishnan, A., et al. (2023). [Measuring Faithfulness in Chain-of-Thought Reasoning](#). arXiv:2307.13702.
- [7] Alioto, J. P. (2026, March 16). [Commentary on CoT as a token-level control surface](#). X.
- [8] Chen, Y., Benton, J., Radhakrishnan, A., et al. (2025). [Reasoning Models Don't Always Say What They Think](#). arXiv:2505.05410. Version 1, May 8.
- [9] Korbak, T., Balesni, M., Barnes, E., et al. (2025). [Chain of Thought Monitorability: A New and Fragile Opportunity for AI Safety](#). arXiv:2507.11473. Version 1, July 15.
- [10] Baker, B., Huizinga, J., Gao, L., et al. (2025). [Monitoring Reasoning Models for Misbehavior and the Risks of Promoting Obfuscation](#). arXiv:2503.11926.
- [11] Lindsey, J. (2025). [Emergent Introspective Awareness in Large Language Models](#). *Transformer Circuits*, October 29; revised January 1, 2026. arXiv:2601.01828.
- [12] Vogel, T. (2025, December 12). [Small Models Can Introspect, Too](#). Experimental article and accompanying code.
- [13] Alioto, J. P. (2026). *Cognitive Kernel Networks: Geometric Dominance and Preservation of Computation*. Protocol v1.4.2, Release Candidate, September 5.